

Performance Comparison of Machine Learning Models in Phishing Email Classification

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Abstract—Phishing emails remain a major cybersecurity threat because they deceive users into revealing sensitive information through messages that often resemble legitimate communication. This study compares five machine learning models for phishing email classification using the CEAS08 dataset from Kaggle. The selected attributes include the email body and label, while an additional body_length feature was engineered to capture structural information. Email text was transformed into numerical representations using TF-IDF with the top 5,000 terms, then combined with body_length to form the final feature matrix. The dataset was split into training and testing sets using stratified sampling, and SMOTE was applied only to the training set to address class imbalance. Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Naive Bayes were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Random Forest achieved the best overall performance, demonstrating strong effectiveness for phishing email detection in practical, modern, resource-efficient cybersecurity classification tasks.

Index Terms—Phishing email detection, Machine learning, Random Forest, Support Vector Machine, Naive Bayes, Logistic Regression, Decision Tree, classification

I. INTRODUCTION

Phishing attacks remain one of the most persistent cybersecurity threats because they exploit human trust rather than technical weaknesses alone. Through deceptive emails, attackers attempt to persuade users to reveal sensitive information such as login credentials, financial data, or personal information. This problem is particularly challenging because phishing emails are often designed to imitate legitimate communication, making them difficult for users to identify manually. As a result, phishing email detection has become an important research topic in cybersecurity, especially with the increasing use of machine learning and natural language processing approaches to improve detection performance [1], [2].

In real-world conditions, the impact of phishing continues to increase as attacks become more adaptive and difficult to distinguish from normal communication. Recent reports show that phishing remains a large-scale threat, with millions of phishing attacks recorded around the world and significant financial losses caused by business email compromise [1]. Traditional detection approaches such as blacklists, whitelists, and rule-based filtering are often insufficient because attackers can modify email content, sender information, and URLs to

avoid detection [3], [4]. Therefore, a more adaptive and data-driven approach is needed to classify phishing emails before users interact with malicious content [5].

To address this issue, this study compares the performance of five classical machine learning models for phishing email classification: Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Naive Bayes. The CEAS08 phishing email dataset is used as data source, with email body text transformed using TF-IDF and email length added as a structural feature. The models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to determine which method provides the most effective performance. By focusing on traditional machine learning models, this study aims to provide a practical, interpretable, and computationally efficient approach for phishing email detection.

II. LITERATURE REVIEW

Phishing email detection has become an important research topic because phishing attacks continue to evolve and remain difficult to identify using conventional methods. Review studies have shown that machine learning, natural language processing, and automated classification techniques are increasingly used to improve phishing email detection performance [1], [2]. Traditional approaches such as blacklists, whitelists, and rule-based filtering are still useful, but they often struggle when phishing emails become more adaptive, deceptive, and similar to legitimate communication [3]. Comparative studies also show that machine learning can provide a practical solution for detecting phishing emails before users interact with malicious content [4].

Previous studies have also emphasized the importance of feature representation in phishing email classification. Deep learning has been explored in phishing detection, but it often requires larger datasets and higher computational resources, making classical machine learning still relevant for practical implementation [5]. Text-based feature extraction methods such as TF-IDF, word embedding, FastText, and related NLP techniques have been used to transform email content into numerical representations that can be processed by classification models [6], [7]. Broader studies further confirm that the quality of feature extraction strongly influences model performance in phishing detection tasks [8]. Classical machine learning

models remain widely used because they are efficient, interpretable, and suitable for comparative classification studies. Prior research comparing Support Vector Machine, Random Forest, Logistic Regression, Decision Tree, and Naive Bayes has shown that these models can achieve strong performance on phishing email datasets [9]. In particular, Random Forest and Support Vector Machine are frequently reported as competitive classifiers, although their performance depends on the dataset and feature representation used [10]. Other comparative studies also support the usefulness of traditional machine learning, especially when the goal is to obtain reliable performance with lower computational complexity [11].

Although this study focuses on classical machine learning, related works on phishing and spam detection provide useful context for understanding current research directions. Some studies have reviewed machine learning techniques for phishing detection and highlighted the importance of selecting suitable features and evaluation metrics [12]. Other works have applied deep learning for spam and phishing email filtering, showing promising results but also requiring more complex model development [13]. In addition, URL-based phishing detection studies indicate that features beyond email body text, such as malicious links and hyperlink patterns, may improve detection performance [14].

Recent phishing detection studies have also explored explainable, AI-assisted, and transformer-based approaches to improve model transparency and robustness [15], [16]. Other studies have examined phishing behavior from broader perspectives, including phishing email characteristics, hyperlink profiling, and URL-based activity analysis [17], [18]. Comparative research using TF-IDF, Word2Vec, and BERT further shows that different feature representation methods can influence classification results [19]. Based on these findings, this study focuses on a practical comparison of classical machine learning models using TF-IDF-based email body representation and email length as an additional structural feature.

III. METHODOLOGY

This subsection outlines the general design of the research study. The workflow includes data acquisition, data preprocessing, feature engineering, exploratory data analysis, data balancing and partitioning, model development, and model evaluation. Each stage was arranged sequentially to ensure that the raw dataset could be transformed into a machine-readable format before being used for classification. After the dataset was obtained, the relevant features were selected and prepared through preprocessing. The email body was transformed into numerical features using TF-IDF vectorization, while the `body_length` attribute was added as a structural feature. The dataset was then divided into training and testing sets using stratified sampling, and SMOTE was applied only to the training data to address class imbalance. The trained models were evaluated using the testing set to ensure that the performance measurement reflected unseen data. Several evaluation metrics, including accuracy, precision,

recall, F1-score, and ROC-AUC, were used to compare the effectiveness of each model. Fig. 1 illustrates the complete research workflow. Each step of the workflow is described in more detail in Subsection III-A to Subsection III-G.

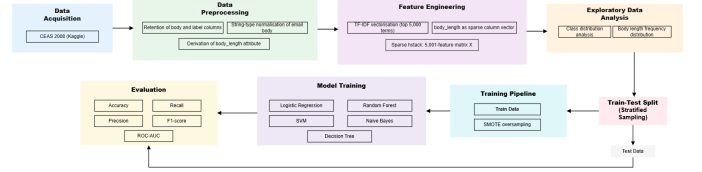


Fig. 1. Research workflow of email phishing prediction framework.

A. Data Description

The dataset used in this study is the CEAS 2008 Phishing Email [20] from Kaggle [21], a publicly available corpus collected for phishing email research. This dataset contains email records with both phishing and legitimate classes, originally compiled from various real email sources submitted to the CEAS (2008 Conference on Email and Anti-Spam) challenge.

Each email entry includes metadata attributes such as sender, receiver, date, subject, body text, associated URLs, and a binary target label indicating whether the email is phishing (1) or legitimate (0). The dataset was obtained in CSV format.

In this study, the focus was on the core textual content and classification label. Therefore, only the email body text and the corresponding label were selected for model development. Other fields such as sender, receiver, and subject were preserved for potential future feature engineering. The feature descriptions, types, and units are summarized in Table I.

TABLE I
FEATURE DESCRIPTIONS, TYPES, AND MEASUREMENT UNITS.

Feature	Description	Type
body	Full textual content of the email	Text
label	Target class of the email (0 = Legitimate, 1 = Phishing)	Categorical
body_length	Number of characters in the email body	Numeric

B. Data Preprocessing

Data preprocessing was performed to prepare the CEAS08 dataset for machine learning analysis. Only the `body` and `label` columns were retained from the original dataset. Email bodies were cast to string format to ensure consistent text handling and avoid processing errors during execution.

To support both exploratory analysis and feature construction, a numerical attribute representing email length in characters (`body_length`) was computed for every email. The target label (`y`) was set as the `label` column, with 0 indicating legitimate and 1 indicating phishing emails.

C. Feature Engineering and Transformation

Feature engineering involved constructing the final feature matrix X by combining two complementary representations. First, the email body text was vectorized using Term

Frequency-Inverse Document Frequency (TF-IDF), with English stop words removed and the vocabulary restricted to the top 5,000 most informative terms, yielding a sparse text feature matrix. Second, the numerical `body_length` attribute was converted into a sparse column vector. Both representations were then horizontally concatenated using sparse matrix stacking (`hstack`), producing a final feature matrix X of 5,001 features that captures both the textual content and structural length of each email.

D. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the CEAS08 phishing email dataset. The label distribution was visualized using a bar chart to examine the proportion of legitimate and phishing emails. In addition, email body length was analyzed using a histogram with 50 bins over the range of 0 to 35,000 characters. The histogram shows that most emails are relatively short, while a small number of emails have considerably longer content. These findings supported the inclusion of `body_length` as an additional feature and guided the preprocessing decisions for model training.

The analysis also indicates that the data set has two target classes, namely, Legitimate and Phishing emails. There is a little class imbalance, with the majority class being Phishing emails. The imbalance is not too high, but it should be taken into account when training the model to minimize the risk of making biased predictions towards the dominant class. Hence, besides accuracy, the accuracy evaluation metrics like Precision, Recall, and F1-score were taken into consideration for a comprehensive evaluation of the model performance.

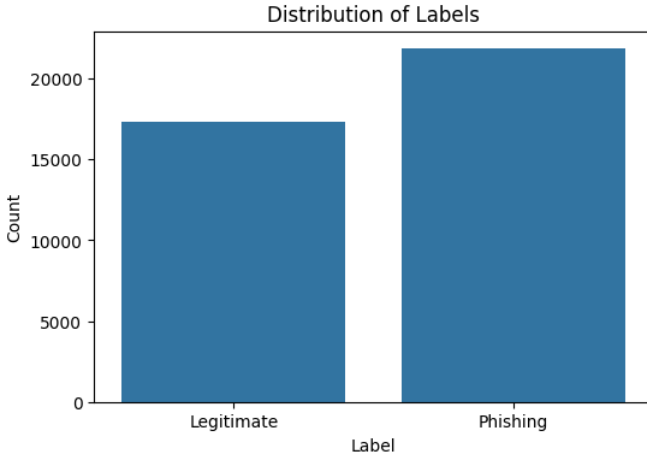


Fig. 2. Distribution of Email Classes in the Dataset

The email length distribution, as depicted in Figure 3, exhibits a pronounced right-skewed pattern, indicating that the vast majority of emails in the dataset are relatively short in character length. The presence of extreme outliers in email length necessitates the application of text truncation or normalization strategies to ensure that excessively long documents

do not disproportionately influence the learned representations during model training. Furthermore, the dominance of short emails in the dataset highlights that email length may serve as a useful discriminative feature in the classification pipeline, given that its distribution likely differs between the Legitimate and Phishing classes.

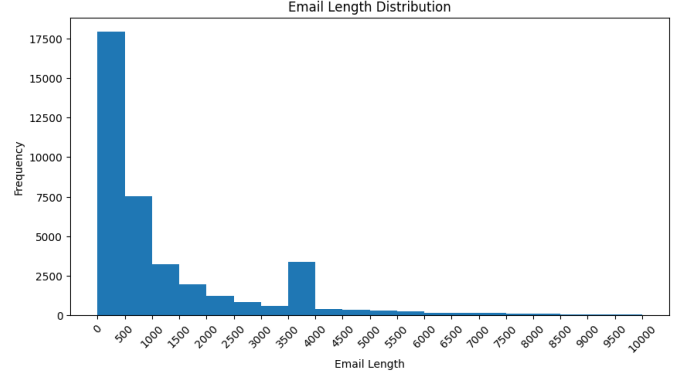


Fig. 3. Distribution of Email Length Across the Dataset

E. Data Balancing and Partitioning

The dataset was divided into training and testing sets using stratified sampling (80% training, 20% testing) to preserve class proportions across both splits. SMOTE was then applied exclusively to the training set to address class imbalance by synthetically generating minority class samples, while the test set remained unaltered to ensure unbiased evaluation.

F. Model Development

To identify phishing emails, five machine learning algorithms have been chosen. The logistic regression and support vector machine models are used for email detection. Logistic regression and support vector machine models are used for email detection. (SVM), Random Forest, Decision Tree and Naive Bayes. They are all examples of different learning styles, such as: Based on the selected method, the classification can be linear, margin-based, ensemble, tree-based or probabilistic approaches, which make it possible to make a comprehensive comparison. Each The training set considered for the training of model was sampled using SMOTE. The algorithm was not able to deal with sparse matrices, which is why Naive Bayes The feature matrix needed to be converted to a dense array, a process that was required. prior to training. The key hyperparameters for each model are summarized in Table II.

TABLE II
KEY HYPERPARAMETERS OF THE EVALUATED MACHINE LEARNING MODELS.

Model	Key Hyperparameters
Logistic Regression	max_iter=1000
SVM	kernel=linear, probability=True
Random Forest	n_estimators=200, random_state=42
Decision Tree	random_state=42
Naive Bayes	— (default)

G. Model Evaluation

Model performance was assessed using five metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. ROC-AUC was designated as the primary metric, as it evaluates model performance across all classification thresholds rather than at a single decision boundary, making it a more robust indicator of overall discriminative capability in phishing detection contexts. A confusion matrix was generated for the best-performing model to provide a detailed breakdown of misclassification patterns between phishing and legitimate classes. Additionally, ROC curves were plotted for all models in a single figure with the AUC value annotated for each, enabling direct visual comparison of their discriminative ability.

IV. RESULT AND DISCUSSION

A. Impact of Feature Engineering on Model Performance

Table III presents the performance comparison of the machine learning models before incorporating the email length feature. In this condition, the models were evaluated using the original TF-IDF representation of the email body. Random Forest achieved the best overall performance, with the highest Accuracy (0.99374), Recall (0.99565), F1-score (0.99440), and ROC-AUC (0.99976). This indicates that Random Forest was highly effective in distinguishing phishing emails from legitimate emails even without the additional structural feature. SVM and Logistic Regression also produced strong results, with both models achieving ROC-AUC values above 0.999. Meanwhile, Decision Tree and Naive Bayes showed slightly lower performance compared to the other models, although their results were still relatively high.

TABLE III
PERFORMANCE OF MACHINE LEARNING MODELS BEFORE
INCORPORATING EMAIL LENGTH FEATURE.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.99374	0.99315	0.99565	0.99440	0.99976
SVM	0.99208	0.99517	0.99062	0.99288	0.99942
Log. Regression	0.99183	0.99426	0.99107	0.99266	0.99931
Decision Tree	0.98595	0.98387	0.99107	0.98746	0.98707
Naive Bayes	0.98493	0.99465	0.97871	0.98638	0.98592

Table IV shows the model performance after incorporating the email length feature into the classification process. After adding this feature, Random Forest remained the best-performing model, achieving the highest Accuracy of 0.99451, Recall of 0.99519, F1-score of 0.99508, and ROC-AUC of 0.99972. Compared to the previous result, Random Forest showed a slight improvement in Accuracy and F1-score, suggesting that email length provided additional useful information for classification. Logistic Regression and Naive Bayes also experienced small improvements in several metrics, while SVM remained nearly unchanged. However, Decision Tree showed a slight decrease after adding the feature, indicating that the additional feature did not benefit all models equally. Overall, the results suggest that email length can support

model performance, but the main predictive power still comes from the TF-IDF representation of the email body.

TABLE IV
PERFORMANCE OF MACHINE LEARNING MODELS AFTER
INCORPORATING EMAIL LENGTH FEATURE.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.99451	0.99497	0.99519	0.99508	0.99972
SVM	0.99208	0.99517	0.99062	0.99289	0.99943
Log. Regression	0.99208	0.99426	0.99153	0.99290	0.99931
Decision Tree	0.98557	0.98386	0.99039	0.98711	0.98644
Naive Bayes	0.98519	0.99465	0.97871	0.98662	0.98615

B. Machine Learning Model Performance Comparison

The performance comparison of the five machine learning models is presented in Table IV. The evaluated models include Random Forest, SVM, Logistic Regression, Decision Tree, and Naive Bayes. Each model was assessed using accuracy, precision, recall, F1-score, and ROC-AUC to compare their effectiveness in detecting phishing emails.

Based on Table IV, Random Forest achieved the strongest performance across most evaluation metrics, with the highest accuracy, recall, F1-score, and ROC-AUC. This indicates that Random Forest was effective in identifying phishing emails while maintaining a balanced classification performance. SVM obtained the highest precision score, suggesting that it produced fewer false positive predictions. However, its recall and F1-score were slightly lower than Random Forest. Logistic Regression also showed competitive performance, while Decision Tree and Naive Bayes achieved lower results compared to the other models.

To further evaluate the classification ability of each model, ROC curve analysis was conducted, as shown in Fig. 4. The ROC curve illustrates the relationship between the true positive rate and false positive rate across different classification thresholds.

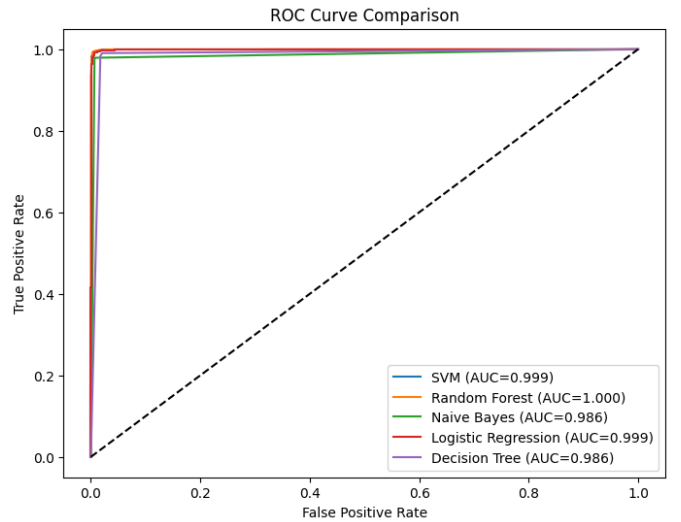


Fig. 4. ROC curve comparison of machine learning models.

The ROC curve supports the results shown in Table IV. The model that yielded the best ROC-AUC was dom Forest with a score of 0.99972, The second one is SVM with 0.99943 followed by Logistic Regression with 0.99931. Decision Tree and Naive Bayes obtained lower ROC-AUC values, which reflect a lower discrimination ability. compared to the top-performing models. Therefore, Random Forest was chosen as the top model phishing. In this study, the detection was carried out using an email.

C. Best Model Analysis

Based on the performance comparison, Random Forest was selected as the best-performing model in this study. The model achieved an accuracy of 0.99451, recall of 0.99519, F1-score of 0.99508, and ROC-AUC of 0.99972. These results indicate that Random Forest was highly effective in classifying phishing and legitimate emails.

The high recall score shows that Random Forest was able to detect most phishing emails correctly. This is important in phishing detection because undetected phishing emails may lead to security risks for users. In addition, the high F1-score indicates that the model maintained a strong balance between precision and recall, meaning that it could detect phishing emails accurately while keeping incorrect predictions low.

The confusion matrix in Fig. 5 further supports this result. Random Forest produced only a small number of misclassifications, with relatively few false positives and false negatives. This indicates that the model was able to classify both legitimate and phishing emails consistently.

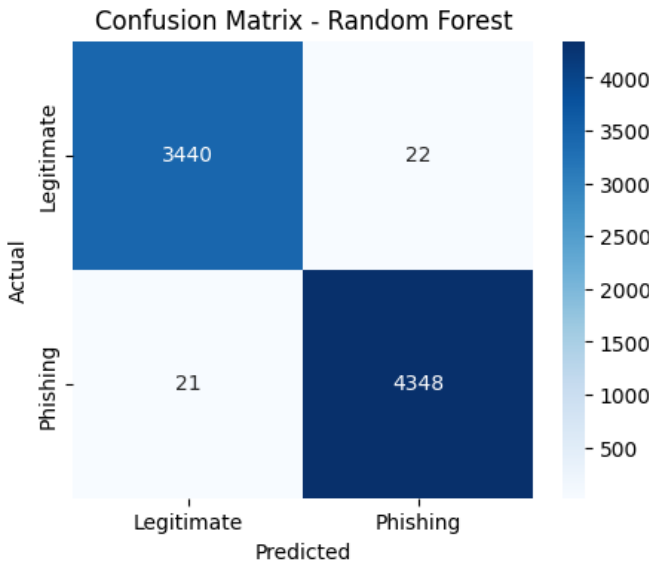


Fig. 5. Confusion matrix of the Random Forest model.

The ROC-AUC score is 0.99972, which is also confirmed. model's strong discriminative ability. Random means that the number comes out of the box at random. Forest had a great ability to distinguish phishing and non-phishing emails. The classification thresholds for legitimate emails, across different

categories. Thus, the Random Forest model was thought to be the most appropriate, This study employs the model for the detection of phishing email.

V. CONCLUSION

This study aims to compare the performance of five machine learning models for phishing email classification based on the CEAS08 phishing email data set. The body of the email was vectorized by TF-IDF and the length of the email was included as a structural feature for classification. To tackle the imbalanced classes, SMOTE was used in the training set. Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Naive Bayes were some of the models evaluated. According to the experimental results, the accuracy, recall, F1-score and ROC-AUC of Random Forest were 0.99451, 0.99519, 0.99508 and 0.99972, respectively, which are the maximum among all the models. By means of the confusion matrix and ROC curve, the other models were once again shown to be effective in separating the phishing emails from the legitimate emails, with Random Forest performing well.

There are some limitations to this study, however. The experiment was conducted on only one data set, namely CEAS08, which can result in a limited generalizability of the model to newer phishing email patterns. Further, only email body text and email length were explored and used in the feature set: sender info, subject line, urls and temporal features were not explored and used. Future research will employ updated and numerous sets of emails, add more features from email content and URLs, and compare the performance of phishing email detection against deep learning or transformer-based models.

AUTHORS CONTRIBUTION

Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing: Nickholas Steven Sukmadjaja. Support in manuscript writing and revision was provided by Nicholaus Hendrik Jeremy and Azani Cempaka Sari. All authors read and approved the manuscript.

DATA AND MATERIALS AVAILABILITY

This study utilized data accessible at <https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset>

AI USAGE DECLARATION

We declare that we use generative tools to enhance the quality control of manuscript writing. No part of the manuscript is part of generative tool, and all the result from generative tool has been modified under human supervision.

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